

Week 1 Technical Handout

Activated oxime E/Z structure generation, AIMNet preoptimization, and classical Beckmann-rule benchmark

Input: attached substrate list with Product A/Product B ratios and conversion notes. Use these data as the experimental benchmark column.

Goal

For every ketone substrate, generate the corresponding **activated oxime** model, make both **E** and **Z** isomers, preoptimize each with AIMNet, identify the lower-energy isomer, measure the stereochemical dihedral, and compare the classical Beckmann-rule prediction with experiment.

1. Required atom mapping and chemical transformation

Original ketone fragment	Activated oxime model to build	Required atom labels
C(aryl side)-C(=O)-C(allyl side)	C(aryl side)-C(=N-OH ₂ ⁺)-C(allyl side)	C_aryl, C_oxime, N_oxime, O_oxime, C_allyl

Important: preserve the atom map for the studied oxime fragment in every molecule. Only replace the ketone O by the N-OH₂⁺ activated oxime group.

2. Structures to generate and optimize

1	Create the 3D ketone structure from the attached substrate list or provided SDF/SMILES.
2	Replace the ketone C=O with C=N-OH ₂ ⁺ , keeping C_aryl-C_oxime-C_allyl atom numbering fixed.
3	Generate two isomers per molecule: E and Z . Use CIP priority only for naming E/Z.
4	Preoptimize E and Z with AIMNet. Save initial and optimized XYZ/SDF files.
5	Extract E _{min} (E), E _{min} (Z), and identify the lower-energy isomer.
6	For the lower-energy isomer, measure O_oxime-N_oxime-C_oxime-C_aryl and optionally O_oxime-N_oxime-C_oxime-C_allyl.
7	Apply the classical Beckmann rule: the group anti/trans to the activated N-O leaving group is predicted to migrate.

3. Output folder convention

week1_activated_oximes/mol_001/mol_001_E.xyz, mol_001_Z.xyz, mol_001_E_opt.xyz, mol_001_Z_opt.xyz, mol_001_E.log, mol_001_Z.log

4. Interpretation rules

Measured geometry in lower-energy isomer	Meaning	Classical Beckmann prediction
O-N-C_oxime-C_aryl approx. 180 degrees	Aryl side is anti to activated N-O leaving group	Aryl-side migration
O-N-C_oxime-C_allyl approx. 180 degrees	Allyl/alkyl side is anti to activated N-O leaving group	Allyl/alkyl-side migration
Dihedral not close to 0 or 180 degrees	Geometry may be twisted/nonideal after preoptimization	Flag for manual inspection before using prediction

5. Week 1 deliverable

Submit: (i) optimized E/Z structures for all substrates, (ii) one CSV summary table, and (iii) a short note identifying which substrates agree or disagree with experiment and which require manual inspection.

Week 1 Summary Table Template

Fill one row per substrate. Add extra rows in the CSV version.

Mol ID	Structure / file	C_aryl	C_ox	N_ox	O_ox	C_allyl	Emin(E)	Emin(Z)	Delta E Z-E	Lowest isomer	Dihedral O-N-C-Ar	Dihedral O-N-C-Allyl	Anti group	Beckmann pred.	Exp. A/B	Agreement / notes
mol_001	mol_001.png / xyz	—	—	—	—	—	—	—	—	E or Z	— deg	— deg	aryl/allyl	A or B	33/67 etc.	yes/no/unclear

Manual checklist before accepting each row

Ketone converted to C=N-OH2+; atom map preserved; E and Z both generated; AIMNet finished; energies extracted; lower-energy isomer identified; anti group assigned from geometry; prediction compared to attached experimental A/B ratio; problematic geometry flagged.